

LENGTH: say which spaces count

Check the value behind the display

01 / THE SAS EXAMPLE

```
data test;
  length text $10;
  input text $;
  len = length(text);
  datalines;
ABC
ABC
ABCDEFGHIJ
X
.
;
run;

proc print data=test noobs;
run;
```

REPORTED TEST SETUP / 21 SEP 2026
Claude Sonnet 4.5 / AWS Bedrock GovCloud
Claude Code 2.1.278 / effortLevel=high / prompt v0.1.0

ATTACH SOURCE OR PASTE IT AFTER THE PROMPT. START FRESH EACH TIME.

WHAT YOU WILL PRACTICE

Specify an invisible data detail and ask for a check that makes it visible.

Each text value occupies ten character positions. LENGTH ignores trailing blanks and returns 1 for all blanks. The period on the final input row supplies a missing character value.

EXPECTED RESULT

Row	Stored width	len
1	10	3
2	10	3
3	10	10
4	10	1
5	10	1

Both ABC rows contain seven padding spaces.
Final row: ten spaces; LENGTH returns 1.

REPORTED NUMERIC MATCHES: A 3/3 | B 3/3
Required behavior: A 0/3 | B 3/3

Reference values checked in Python; SAS execution pending.

Prompt A: leave choices open

High degrees of freedom / Notice the decisions the model makes for you.

COPY THIS PROMPT + THE SAS EXAMPLE

```
Convert this SAS code to Python.  
  
Show code in this response. Do not create or run files,  
or read existing Python files.
```

REPORTED TRIAL RESULTS / 21 SEP 2026

Matching numbers; missing behavior

Run	Numeric values	Required behavior
A1	Yes	No
A2	Yes	No
A3	Yes	No

All three used pandas .str.len() directly.
Reported: no padding; final period kept as literal '.'.
The numeric vector [3, 3, 10, 1, 1] matched, but stored text and missing-value interpretation did not.

SOURCES: results-20260921.csv; trial-log-20260921.md; analysis-20260921.md

WHY CHOICES REMAIN OPEN

- 1

The numbers can agree
Literal '.' has length 1, which matches the expected blank result by coincidence.
- 2

Storage still differs
This fixture requires width 10. Correct numeric values alone do not establish stored-text equivalence.
- 3

Name the spaces to ignore
For these ASCII strings, ignore trailing ordinary spaces in the calculation; preserve the stored text.

COUNTEREXAMPLES / LOCALLY CHECKED

Python value	Direct len	Expected LENGTH
'ABC '	6	3
"	0	1
' ABC '	5	4

ANTI-PATTERN: CHECKING ONLY THE NUMBERS

Inspect stored values, missing input and trailing spaces, even when the numeric results agree.

Prompt B: make the choices explicit

Low degrees of freedom / State the behavior, then ask for a check.

COPY THIS PROMPT + THE SAME SAS EXAMPLE

```
Convert this SAS code to Python using pandas.
- Create text from ABC, ABC, ABCDEFGHIJ, X, and an empty string.
  The final SAS input period represents character missing here.
- Right-pad every text value with spaces to width 10.
- Preserve the stored text. For len, remove trailing ordinary
  spaces only with .str.rstrip(' '), then count characters.
- Match SAS LENGTH for a blank string: set its result to 1.
  Use .str.len().clip(lower=1) after the trailing-space removal.
- Print a visible representation of text and the len column.
- Include assertions: stored widths are all 10; len values are
  [3, 3, 10, 1, 1] in the original row order.
- Briefly explain why stored width and LENGTH can differ.
```

Show code in this response. Do not create or run files, or read existing Python files.

WHY THESE DETAILS HELP

- 1 Make the stored value visible**
Right-pad to width 10 and inspect repr(text). Interpret the final input period as character missing.
- 2 Preserve the counting rule**
Use rstrip(' ') for the calculation only. Keep leading spaces; set the all-blank result to 1.
- 3 Verify the requested behavior**
B1-B3 report width and value assertions passing. Correct padding can use different methods.

REPORTED RESULTS / THREE B RUNS

Run	Numeric values	Required behavior
B1	Yes	Yes
B2	Yes	Yes
B3	Yes	Yes

CHECK IT YOURSELF
Confirm all widths = 10 and the five len values.
Final stored value: ten spaces, not a literal ' '.

SAS REFERENCE

Reports describe six fresh parallel agents; complete transcripts are absent from this repository.
Passing assertions are reported. Local reference checked; SAS run pending. Three runs do not guarantee future results.

One behavior, three implementations

One behavior contract, three implementations. Run the companion files and the same three acceptance tests against shared fixture data.

SAS / SOURCE BEHAVIOR

```
%macro translate(in=, out=);
data &out;
  length text $10;
  set &in;
  len = length(text);
run;
%mend;
```

SAS LENGTH ignores trailing ordinary blanks and returns 1 for an all-blank character value; LENGTH is byte-based.

PYTHON / PANDAS

```
def translate(df):
    out = df.copy(deep=True)
    out["text"] = out["text"].str.ljust(10)
    trimmed = out["text"].str.rstrip(" ")
    out["len"] = trimmed.str.len().clip(
        lower=1)
    return out
```

Pad stored text to 10. Remove only ordinary trailing spaces for len, then enforce a minimum of 1; keep stored text intact.

R / BASE

```
translate <- function(df) {
  out <- df
  out$text <- sprintf("%-10s", out$text)
  s <- sub(" +$", "", out$text)
  out$len <- pmax(nchar(s), 1L)
  out
}
```

The base R pattern " +\$" removes trailing ordinary spaces only. nchar counts the remaining ASCII characters; pmax enforces 1.

Trailing spaces only

Leading spaces and tabs remain meaningful. Broad whitespace trimming changes the rule even if the original numeric results happen to match.

Length differs from width

Every stored value has width 10. LENGTH counts through the final nonblank character and returns 1 for the all-blank value.

SAS LENGTH

pandas rstrip

R trim rules

Three checks that earn the pass

Use the same inputs and expected behavior in each language. Check the contract as well as the values.

T1 / Original five-row fixture

INPUT / CONDITION

ABC, ABC, ABCDEFGHIJ, X and character missing decoded as an empty string.

EXPECTED

len = [3, 3, 10, 1, 1]; every stored text value is padded to width 10.

WHAT THIS CATCHES

Counting storage padding, losing character-missing blanks, or returning zero for an all-blank value.

T2 / Added meaningful-space boundary

INPUT / CONDITION

Added validation strings " ABC ", "ABC<tab>" and three spaces; CSV contains the actual tab.

EXPECTED

len = [4, 4, 1]; leading space and tab survive; all-blank result is 1.

WHAT THIS CATCHES

strip() removing leading blanks, broad whitespacerstrip(), or omitting the minimum of 1.

T3 / Preservation, widths and order

INPUT / CONDITION

All shared rows reversed; original caller data saved.

EXPECTED

Caller unchanged; returned text equals expected padded text; integer lengths and row order retained.

WHAT THIS CATCHES

Mutating or trimming stored text, losing padding, wrong result type or row reordering.

REFERENCE CODE + TESTS: [module-03-length/appendix/](#)

Run: `python tools/run_appendix_tests.py --modules 03`

New reference code, separate from historical trials. SAS checks await execution; scope is stated in the companion README.